NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

LINK, METALLIC BELT, 20MM., M8

This specification was approved by the Departments of the Army, Navy, and Air Force for use of procurement services of the respective Departments.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of metallic belt link known as Link, Metallic Belt, 20mm., M8.

2. APPLICABLE SPECIFICATIONS, STANDARD, OTHER PUBLICATIONS, AND DRAWINGS

2.1 Specifications.—The following specifications form a part of this specification:

FEDERAL SPECIFICATION

QQ-M-151—Metals; General Specification for Inspection of.

U. S. ARMY SPECIFICATIONS

100-2—Standard Specification for Marking Shipments by Contractors.[1]

57-0-2—Finishes, Protective, for Iron and Steel Parts.

94-40645—Marking, Exterior, Domestic and Export Shipment by Contractor.[2]

ORDNANCE DEPARTMENT U. S. ARMY TENTATIVE SPECIFICATION

AXS-1759—Compound, Rust Preventive, Soft Film.

(Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.2 Standard.—The following standard forms a part of this specification:

NATIONAL MILITARY ESTABLISHMENT ENGINEERING STANDARD

JAN-STD-105—Sampling Inspection Table for Inspection by Attributes.

(Copies of standards may be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.3 Other publications.—The following publications form a part of this specification:

ARMY ORDNANCE PUBLICATION

AVPS No. 19—Packaging Instructions Pertaining to Links, Disintegrating Belt, 20 mm., M7 & M8.

(Copies of publications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook[3]

(Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.)

2.4 Drawing.—The following drawing forms a part of this specification:

ORDNANCE DEPARTMENT DRAWING

D7230430—Link, Disintegrating Belt, 20 mm., M8.

(Copies of drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

2.5 All drawings, specifications, standards, and publications shall be of the issue in effect on date of invitation for bids. The issue in effect shall be that issue which the contracting officer shall designate or furnish to the bidder.

[1] Applicable only to Army purchases.
[2] Applicable only to Air Force shipments.
[3] Applicable only to Navy shipments.

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3. REQUIREMENTS

3.1 Material.—Unless otherwise specified in the invitation for bids and contract or purchase order, the contractor may use steel of such composition and heat treatment as he considers best suited to produce links meeting the requirements of this specification. (See 6.1)

3.1.2 Prior to production, the contractor shall furnish the contracting officer a copy of complete specifications covering the type of steel to be used.

3.1.3 An analysis of each heat of steel used in manufacture of links shall be furnished by the contractor.

3.2 The links shall meet all requirements of the applicable drawing.

3.3 Rust proofing.

3.3.1 After heat treatment the links shall be given a phosphate coating conforming with the requirements of U. S. Army Specification No. 57–0–2, Type II, Class B and to the following requirements:

3.3.1.1 The phosphated links shall be free from scratches and bare spots.

3.3.1.2 The phosphated links shall be coated with compound rust preventive grade 1, Ordnance Department, U. S. Army Tentative Specification AXS–1759.

3.3.1.3 The residual coating of rust preventive compound shall not be in excess of 25 milligrams per link.

3.3.2 The phosphated and rust preventive compound coated link shall withstand 48 hours salt spray test as prescribed in paragraph 4.5.

3.4 Extraction resistance.—Links shall meet the following requirements for extraction resistance when tested in the manner prescribed in paragraph 4.9. A fall of not more than 23 inches shall extract the gage bar from either the single or double loop of the link. A fall of 13 inches shall fail to extract the gage bar from the single loop. A fall of 9 inches shall fail to extract the gage bar from the double loop.

3.5 Pulling strength.—Finished links, when tested as prescribed in paragraph 4.10 shall withstand a load of 125 pounds, without fracture and without permanent set in excess of 0.010 inch.

3.6 Contractor's inspection.—The contractor shall determine and maintain a system of inspection including the necessary gages and testing equipment adequate to insure links that conform to requirements in this specification.

3.7 The link defects shall be classed as shown in table I. The acceptable quality level shall be 3% for major defectives and 6% for major and minor defectives combined.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified, links shall be inspected and tested at the place of manufacture.

4.2 Lot and sublot size shall be as specified or authorized by the contracting officer.

4.3 Sampling.—Samples for inspection and tests in 4.6 to 4.10, inclusive, shall be selected in accordance with JAN–STD–105. Inspection level III shall be considered normal inspection level.

4.4 After heat-treatment and prior to phosphating, the inspector shall check links at random for hardness to insure compliance with the applicable drawing. The number of links tested shall be based upon production quantities and previous quality, but shall not be less than three links per hour of operation for each heat-treating unit. In the event that the hardness of the links is not within the required limits, the links in process shall be rejected, subject to removal by the contractor of all links not meeting the requirements.

4.5 Test of rustproofing

4.5.1 Salt spray test shall be conducted as prescribed in Federal Specification QQ–M–151.

4.5.2 The forty-eight (48) hour *salt spray tests* shall be conducted on links selected at random after they have been coated with compound rust preventive. At least five links of each four hours of the phosphate coating process shall be selected for testing. The selected links shall be placed in a new, unused link carton for a period of one hour to absorb any surplus compound rust preventive and then the link shall be subjected to the salt spray test. The results of this test shall not be used as a

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basis of acceptance or rejection of the links already processed, but shall be called to the attention of the contractor. After any unsatisfactory test, the links thereafter processed shall be subject to rejection and final acceptance shall be withheld until satisfactory salt spray tests have been completed. After a series of five consecutive lots have successfully met the salt spray test, the original procedure shall again be resumed. When a majority of the links in any test show any signs of corrosion except at the sharp edges, at the completion of the test, it shall be considered a failure. When one or two links show not more than two areas of corrosion and each such areas does not exceed one-eighth of an inch in diameter or its equivalent, the test shall be acceptable unless such test is a repetition of a similar result in any twenty-four hour period in which case it shall be a failure. Links having scratches or bare spots in the phosphate coating shall not be selected for corrective action; such links shall not be cause for rejection except as provided in paragraph 4.12.2.

4.6 Visual inspection.

4.6.1 Evidence of compound rust preventive in excess of the requirements of paragraph 3.3.1.3 shall be checked by a weight method. Except as provided in paragraph 4.12.2, the results of this test shall not be used as a basis for acceptance or rejection but shall be called to the attention of the contractor for corrective action.

4.6.2 Links having scratches or bare spots in the phosphate coating shall be brought to the attention of the contractor for corrective action, but need not be cause for rejection, except as provided in paragraph 4.12.2.

4.6.3 Occurrence of laminated metal in a link shall be classed as a defect in accordance with table I.

4.6.4 Links having cracks or forming tears shall be brought to the attention of the contractor for corrective action but need not be classed as defective unless the cracks or tears are so located on the link or of such magnitude as to adversely affect the functioning of the link. Cracks at or near the base of the ears are particularly hazardous to function. Cracks on the upper surface of the link at the base of the ear are permitted provided they do not continue through to the underside of the ear. Cracks in the flanges do not affect functioning.

4.7 Burrs which affect the functioning of the gun, scratch the cartridge case or feeding components, or injure personnel in handling are not acceptable and shall be cause for rejection. Occurrence of such burrs on a link shall be classed as a defect in accordance with table I.

4.8 Inspection for dimensions.—

4.8.1 Gaging for dimensions shall precede any test which can alter the link dimensions *i.e.* extraction test.

4.8.2 In gaging the single and/or double loop diameters, the link shall be considered acceptable if, acting under its own weight only, it pivots freely about or drops off the go portions, and fails to so pivot or drop off the no-go portions of the required gage members.

4.8.3 Wherever applicable, the ability of a link acting under its own weight only to pivot about or drop out of a gage shall be used as the standard for determining conformity with the gage.

4.9 Extraction test.—Finished links shall be tested for required extraction resistance by use of a hardened and ground steel bar or plug in lieu of a cartridge. The bar shall be 0.953 inch in diameter, and shall support a pendent rod at right angles to the axis of the bar at the midpoint of its length. The rod shall be provided with a freely sliding drop block of one pound weight, and a stop at the lower end of the rod so positioned as to permit the block to fall freely through the required distance. Total weight of the bar, pendent rod, and stop, exclusive of the drop block, shall be 1.25 pounds. With the bar inserted in the loop of the link so that the rod hangs below the center of width of the link, and the link supported in horizontal position by means of the ears and the flange of the single loop, the drop block shall be permitted to fall freely through the prescribed distance and strike the stop. Each link, when tested as above, that does not meet the requirements of paragraph 3.4. shall be classed as defective in accordance with table I.

4.10 Pulling test.—Links shall be subjected to the following test: The link shall be assembled on two parallel studs of such diameter

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as to fit easily into the loops of the links. The studs shall then be forced apart laterally to produce the pull specified in paragraph 3.5. Under this test, the link shall not show fracture or take permanent set in excess of the amount specified in paragraph 3.5. Each link tested as above and failing to meet the requirements of paragraph 3.5 shall be classed as defective in accordance with table I.

4.11 Functioning test.—On initial contracts, 100 links selected at random from the first lot shall be forwarded as directed by the contracting officer, to Watervliet Arsenal, Watervliet, N. Y., for functioning test and approval prior to quantity production.

4.12 Rejections.

4.12.1 When a sublot of links submitted by the contractor has more than the allowable number of defectives in the acceptance sample, the entire sublot shall be rejected for reinspection and removal of all defective links before resubmission to the Government inspector. Failure of a sublot on resubmission shall be cause for final rejection of the entire sublot.

4.12.2 Items of inspection described in paragraphs 4.3, 4.5, 4.6.1, 4.6.2, and 4.6.4 not normally used as a basis for acceptance or rejection shall be subject to corrective action. Failure on the part of the contractor to take corrective action at a time satisfactory to the Government inspector shall be sufficient cause for the Government inspector to reject all sublots containing such defective links processed thereafter.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing shall conform to applicable instructions and requirements contained in AVPS No. 19.

5.2 Shipping containers.—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification No. 100–2; for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

6. NOTES

6.1 Steel equivalent to WD 1050 has been used successfully for the manufacture of these links, but the contractor is not obligated to use steel of this specification.

6.2 The classification of defect as minor B, minor A, or major, is, in general, determined by the degree of defect rather than by the nature of the defect. See table I for tabulation of defects.

6.2.1 A minor B defect is one of such degree as to have no effect on end requirements such as functioning, interchangeability or life of the link.

6.2.2 A minor A defect is one of such degree as may result in functioning with slightly reduced effectiveness or be of questionable merit especially when present in large quantities.

6.2.3 A major defect is one of such degree as usually result in failure of function by the link or by related items such as feeder and gun.

6.3 Notice.—This specification, together with the specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced, either wholly or in part, except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

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TABLE I.—*Classification of dimensional defects*[1]*—20 mm., M8 link.*

Column I Drawing dimension or type of test	Column II Limits of minor B[2] defects	Column III Limits of minor A[3] defects	Column IV Major defects[4]
Height of ear	+0.036	+0.045	Exceeding limits of column III.
0.09+0.03 −0.000	0.09 −.003	0.09 −.007	
Clearance radius	+.024	+.03	Do.
0.630+0.02 −0.000	.630−.002	.630−.005	
I. D. single loop	+.011	+.013	Do.
0.954+0.009 −0.000	.954−.004	.954−.006	
I. D. double loop	+.009	+.011	Do.
0.920+0.007 −0.003	.920−.005	.920−.006	
Center distance—Loops	+.005	+.010	Do.
1.35+0.000 −0.020	1.35 −.025	1.35−.030	
Slot between loops 0.570 min.	.570−.010	.570−.015	Do.
Over-all at double loops 1.43 Max.	1.43 +.010	1.45+.015	Do.
Over-all at ear 1.665 Min.	[6] None	1.665−.005	Do.
Outside radius flange 0.640 Max.	.640+.002	.640+.005	Do.
Inside radius flange 0.565 Min.	.565−.002	.565−.005	Do.
Width single loop 0.540+.005 −.005	Gaging not required [5]		
Extraction test—single loop (specified limits 13 and 23 inches).	[6] None	11″ and 25″	Do.
Extraction test—double loop (specified limits 9 and 23 inches).	[6] None	7″ and 25″	Do.
Pull test			X.
Distorted or nonfunctional links			X.
Burrs which scratch cartridge cases or feeding components.		X	
Burrs which might affect functioning of the gun or injure personnel in handling.			X.
Laminated links			X.
Cracks which affect functioning			X.

[1] "X" in place of dimensions indicates defect is in category indicated at head of column.
[2] Dimensions outside drawing tolerance but within limits of column II constitute minor B defects.
[3] Dimensions outside limits of column II but within limits of column III constitute minor A defects.
[4] Dimensions outside limits of column III constitute major defects.
[5] While routine gaging of this dimension is not required, compliance with drawing tolerance is important for maintaining extraction resistance within the required limits.
[6] All failures to comply with specified limits are classified as minor A or major defects.

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